DATA VISUALIZATION AND ANALYSES FOR TRANSPORT PLANNING

Assessment 3 | BENV7504 Digital Cities | Term 1 2026 | School of Built Environment

MAY 10, 2026

SARFRAJ SHAIK

z5586793 | Term 1 2026 | Master of City Planning

ACKNOWLEDGEMENT OF COUNTRY

I acknowledge the traditional custodian of the land and show my respect for elder's past, present, and emerging through thoughtful and collaborative approaches to our work. I seek to demonstrate my on-going commitment to providing places where Aboriginal people are included socially, culturally, and economically.

Source: sbs.com.au

Table of Contents

List of Figures

List of Tables

1. Objectives and Council-wide Profile and Land Suitability Screening

1.1 Study Objectives

This study proceeds to recognize appropriate locations for a future metro station within the Liverpool Council area, with a particular focus on enhancing connectivity between Liverpool LGA and Western Sydney International Airport and the Aerotropolis. Although, the assignment requires the selection of candidate sites within the chosen LGA. The preliminary screening is a deliberate approach taken at the broader council level as the assignment scenario positions students as part of a local council planning team and strategic transport planning pursues beyond the existing Liverpool urban core. This is especially true after considering the distance of the Western Sydney International airport from the actual LGA. Herein, the airport, the aerotropolis, and numerous major transport corridors and infrastructure projects are established within the wider Liverpool Council area that directly impact the rationale behind metro station positioning.

The specific objectives of this study are:

1. Profile Liverpool Council using suburb-scale demographics, employment, dwelling, and travel patterns from ABS and .id statistical datasets.
2. Identify those suburbs where future population growth, employment concentration and transport demand portray strong potential for metro station establishment.
3. Analyse if transport disadvantage is evident through the relationship between low-income households, public transport usage, car ownership and car-based commutes.
4. Build a land suitability model using entropy-weighted indicators and weighted sum overlay output in ArcGIS Pro
5. Utilise the council-wide suitability map to identify preliminary opportunity areas, prior to filtering down into three candidate station sites through walkability and site comparison analysis.

This way of direction ensures that the final recommendation is not chosen arbitrarily but emerges from an evidence-based procedure.

1.2 Inspection of the Council Area before the LGA/Site Scale

The initial stage of analysis utilizes Liverpool Council as the wider scope of geographic study before narrowing down the assessment to selected LGA-based site selection. This is pivotal considering the issue at the stake is future metro station and not a neighbourhood facility. It is a part of larger Western Parkland City transport problem involving the Western Sydney International Airport, the Aerotropolis, future employment lands, growing suburbs and regional transport corridors.

This assignment briefs asks students to act as a part of planning team at a local council and identify three potential sites for metro station connecting the area to Western Sydney Airport and the Aerotropolis. Thereby, the analysis proceeds at the council level to capture entire strategic geography of Liverpool while remaining critical of airport-oriented transport planning.

Liverpool Council's very own Strategic Planning Statement supports this broader perspective. The LSPS identifies Western Sydney Aerotropolis, Western Sydney International Airport, Liverpool City Centre, growing suburbs, proposed transport corridors, existing public transport routes, the Fifteenth Avenue Smart Transit Corridor and the Bankstown-to-Liverpool metro extension as key elements in their long-term plan.

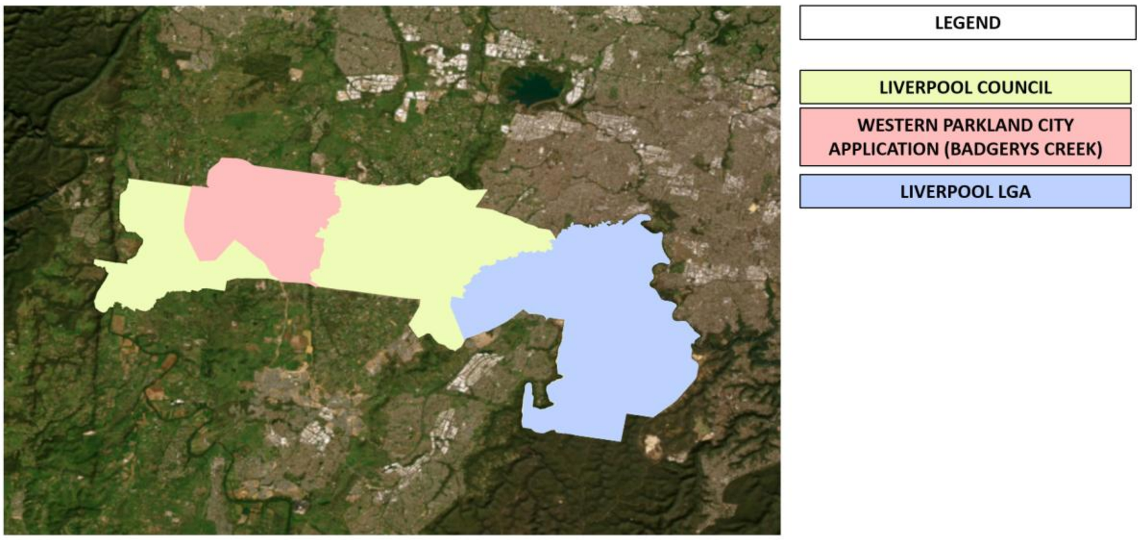

Figure 1Liverpool Council suburb map with LGA and airport location highlighted (made by author on MS Powerpoint)

1.3 Suburb-level Indicators used for Land Suitability Analysis

The council-wide screening approach uses ten suburb-level indicators obtained from ABS and .id datasets. These were selected on basis of their direct impact to the study and desired response to be achieved, as stated in the brief.

Table 1 Reasoning behind the indicators

Indicators	Reasoning
Population density	Existing concentration of residents and potential walk-up demand
Population projection to	
	Future growth pressure and long-term transport demand
Local employment (Labor force aged 15-64)	Employment activity and commuter demand
Forecasted percentage change in dwellings to 2046	Future housing growth and possible TOD demand
Number of people travelling to work by car	Recognizing areas where metro could shift commuters away from private vehicles
Number of people travelling to work by public transport	Existing public transport demand and potential ridership base
Number of households with two or more cars	Embedded car dependency and potential public transport disadvantage
Number of people walking to work	Proximity indicates local employment opportunities, helpful in assessing how active and saturated is the area
Low-income households	Recognizing areas where enhanced public transport can uplift equity and affordability
High-income households	Differentiating higher-income car -owning areas from lower-income transport dependent areas (as seen in forthcoming pages)

This indicator set consciously balances demand, equity and feasibility. Population density and employment show contemporary activity. Population projection and dwelling change display future demand. Car-to-work and two-car households portray car dependency. Public transport and number of people walking to work depict non-car travel behaviour. Low and High-income households help interpret the most beneficial sections of suburbs from availing access to new public transportation.

Table 2 Suburb-level data for all the indicators from ABS and .id

TAG	pop_dens	pop_proj	local_emp	dwell_chng	car_work	pt_work	two_cars	walk_work	low_inc	high_inc
HECKENBERG	3603	8	956	17	486	40	459	5	34	10
BUSBY	3675	5	1219	12	613	56	603	6	31	8
WARWICK FARM	1439	16	2341	24	918	207	581	181	32	7
CHIPPING NORTON	1378	**0**	4193	4	1712	78	581	38	16	27
CARNES HILL	3041	**0**	3432	9	1589	70	1555	27	12	30
HOLSWORTHY	95	11	3183	22	1044	127	937	217	8	38
CASULA	2466	2	6786	11	2929	235	3030	47	21	22
CARTWRIGHT	2843	25	614	40	301	37	291	9	45	5
HINCHINBROOK	3294	**0**	4753	4	2408	105	2254	9	15	23
PLEASURE POINT	714	**0**	1298	1	412	30	545	0	7	52
GREEN VALLEY	3977	**0**	5103	5	2450	136	2405	20	19	21
MILLER	2756	12	798	23	388	45	395	10	45	5
WATTLE GROVE	3466	**0**	4777	7	1779	108	2010	39	11	35
BADGERYS CREEK	59	322	2813	383	1135	16	1429	97	18	27
ASHCROFT	3133	6	1170	13	623	60	593	0	33	10
PRESTONS	1683	**0**	7153	5	3335	190	3103	29	12	31
HAMMONDVILLE	1947	8	1495	5	547	35	667	23	20	23
WEST HOXTON	1635	**0**	4786	1	2230	89	2096	13	11	36
SILVERDALE	0	0	0	0	0	0	0	0	0	0
MOOREBANK	1006	67	5167	90	2006	144	2397	48	15	29
LURNEA	4272	1	2808	5	1418	99	1503	24	30	11
LIVERPOOL	5518	36	11241	52	4727	771	3452	404	28	9
MIDDLETON GRANGE	3047	19	3017	48	1332	44	1452	8	10	32
ELIZABETH HILLS	1745	**0**	1393	2	611	15	643	3	10	33
SADLEIR	3704	1	824	11	412	39	439	9	39	6
CECIL HILLS	1112	**0**	3247	10	1481	37	1430	9	14	35
HORNINGSEA PARK	2645	**0**	1811	5	767	25	782	4	12	34
DENHAM COURT	585	27	954	122	406	20	465	0	21	29
EDMONDSON PARK	2977	63	5867	148	2333	319	2146	34	7	37
AUSTRAL	1106	162	3078	708	1266	130	1250	27	13	24

1.4 Entropy-based Weighting and Weighted Sum Overlay Method

Shannon's entropy weighting method was used to obtain indicators weights prior to performing weighted sum method in ArcGIS pro. This was done to ignore subjective weighting of the indicators. Entropy weighting is useful as it gives higher weight to indicators that show higher variation between suburbs and are more useful in differentiating these spatial units. The method followed a four-step sequence:

1. Raw suburb-level data were compiled for all ten indicators
2. Each indicator was normalised to place all variables on a uniform comparison scale.
3. Shannon's entropy values were enumerated to identify how much variation is noticed in each indicator.
4. Final indicator weights were derived from their diversification and utilised in ArcGIS Pro for weighted sum overlay.

The final entropy-derived weights are as followed:

Table 3 Derived Entropy Weights

Indicator	Entropy-derived weight	Interpretation
Population projection to 2046	0.224	Highest weighted indicator; future population growth is the strongest differentiator
Forecast dwelling change to 2046	0.212	Strong future housing growth is pivotal in long-term station demand
Number of people walking to work	0.150	Indicates the level of localised activities present in the area and pedestrian-access potential
Number of people using public transportation to work	0.103	Captures current public transport demand and potential ridership base

Number of people using car to work	0.063	Shows potential mode-shift opportunity from private vehicles
Local employment (Labor force)	0.059	Depicts employment access and commuter generation
Number of households with two or more cars	0.058	Displays car dependency and possible need for alternative transport
Low-income households	0..056	Need for equity-based public transport
High-income households	0.043	Interprets socio-economic variations in the market demand
Population density	0.030	Lowest weight because 'variation' alone was less decisive than forecast growth

The two most influential indicators were population projection to 2046 and forecasted dwelling percentage change to 2046, with each carrying weights of 0.224 and 0.212 respectively. Together, these two growth indicators account for approximately 43.6% of the overall suitability model. This is highly essential considering metro station is a long-term infrastructure investment and should be located where future growth is strongest rather than wherever current density is high.

Number of people walking to work have also received a relatively high weight of 0.150, portraying that suburbs with stronger local movement patterns may offer better pedestrian-based catchments to the metro station. Number of people using public transport to work gained a weight of 0.103, reflecting upon its existing ridership acting as a base for sustained traffic of the station in both present and future scenarios.

Table 4 Normalised, Proportionate, and Entropy-derived tabular values of the indicators

pop_dens	pop_proj	local_emp	dwell_chn	car_work	pt_work	two_cars	walk_work	low_inc	high_inc
0.6492	0.02481	0.03218	0.02291	0.0418	0.03307	0.05315	0.01238	0.71053	0.15152
0.66239	0.01551	0.05693	0.01556	0.07049	0.05423	0.0987	0.01485	0.63158	0.09091
0.25279	0.04963	0.16251	0.03253	0.1394	0.25397	0.09174	0.44802	0.65789	0.06061
0.24162	0	0.33678	0.00481	0.3188	0.08333	0.09174	0.09406	0.23684	0.66667
0.54625	0	0.26517	0.01132	0.29101	0.07275	0.39987	0.06683	0.13158	0.75758
0.00659	0.03412	0.24174	0.0297	0.16787	0.14815	0.20437	0.53713	0.02632	1
0.44092	0.0062	0.58078	0.01414	0.59376	0.29101	0.8665	0.11634	0.36842	0.51515
0.50998	0.07754	0	0.05516	0	0.0291	0	0.02228	1	0
0.5926	0	0.38948	0.00424	0.47605	0.11905	0.62101	0.02228	0.21053	0.54545
0.11999	0	0.06436	0	0.02508	0.01984	0.08035	0	0	1.42424
0.71771	0	0.42241	0.00566	0.48554	0.16005	0.66878	0.0495	0.31579	0.48485
0.49405	0.03722	0.01731	0.03112	0.01966	0.03968	0.0329	0.02475	1	0
0.62411	0	0.39174	0.00849	0.33394	0.12302	0.54382	0.09653	0.10526	0.90909
0	1	0.20693	0.54031	0.18843	0.00132	0.36001	0.2401	0.28947	0.66667
0.56311	0.01861	0.05232	0.01697	0.07275	0.05952	0.09554	0	0.68421	0.15152
0.29749	0	0.61532	0.00566	0.68549	0.23148	0.88959	0.07178	0.13158	0.78788
0.34585	0.02481	0.0829	0.00566	0.05558	0.02646	0.11895	0.05693	0.34211	0.54545
0.2887	0	0.39258	0	0.43583	0.09788	0.57102	0.03218	0.10526	0.93939
0.17347	0.20782	0.42844	0.12588	0.38522	0.17063	0.66624	0.11881	0.21053	0.72727
0.77175	0.0031	0.20646	0.00566	0.25237	0.11111	0.38342	0.05941	0.60526	0.18182
1	0.11166	1	0.07214	1	1	1	1	0.55263	0.12121
0.54735	0.05893	0.22612	0.06648	0.23294	0.03836	0.36729	0.0198	0.07895	0.81818
0.30885	0	0.0733	0.00141	0.07004	0	0.11136	0.00743	0.07895	0.84848
0.6677	0.0031	0.01976	0.01414	0.02508	0.03175	0.04682	0.02228	0.84211	0.0303
0.19289	0	0.24777	0.01273	0.26661	0.0291	0.36033	0.02228	0.18421	0.90909
0.47371	0	0.11264	0.00566	0.10529	0.01323	0.15533	0.0099	0.13158	0.87879
0.09635	0.08375	0.03199	0.17115	0.02372	0.00661	0.05505	0	0.36842	0.72727
0.53453	0.19541	0.49431	0.20792	0.45911	0.40212	0.58684	0.08416	0	0.9697
0.19179	0.50248	0.23186	1	0.21803	0.15212	0.30339	0.06683	0.15789	0.57576
12.3118	2.45471	7.38412	2.48741	7.4399	3.79894	9.82411	3.31683	10.1579	16.4848

pop_dens	pop_proj	local_emp	dwell_chn	car_work	pt_work	two_cars	walk_work	low_inc	high_inc
0.05273	0.01011	0.00436	0.00921	0.00562	0.0087	0.00541	0.00373	0.06995	0.00919
0.0538	0.00632	0.00771	0.00625	0.00947	0.01428	0.01005	0.00448	0.06218	0.00551
0.02053	0.02022	0.02201	0.01308	0.01874	0.06685	0.00934	0.13507	0.06477	0.00368
0.01963	0	0.04561	0.00193	0.04285	0.02194	0.00934	0.02836	0.02332	0.04044
0.04437	0	0.03591	0.00455	0.03911	0.01915	0.0407	0.02015	0.01295	0.04596
0.00054	0.0139	0.03274	0.01194	0.02256	0.039	0.0208	0.16194	0.00259	0.06066
0.03581	0.00253	0.07865	0.00569	0.07981	0.0766	0.0882	0.03507	0.03627	0.03125
0.04142	0.03159	0	0.02218	0	0.00766	0	0.00672	0.09845	0
0.04813	0	0.05275	0.00171	0.06399	0.03134	0.06321	0.00672	0.02073	0.03309
0.00975	0	0.00872	0	0.00337	0.00522	0.00818	0	0	0.0864
0.05829	0	0.05721	0.00227	0.06526	0.04213	0.06807	0.01493	0.03109	0.02941
0.04013	0.01516	0.00234	0.01251	0.00264	0.01045	0.00335	0.00746	0.09845	0
0.05069	0	0.05305	0.00341	0.04488	0.03238	0.05536	0.0291	0.01036	0.05515
0	0.40738	0.02802	0.21722	0.02533	0.00035	0.03665	0.07239	0.0285	0.04044
0.04574	0.00758	0.00709	0.00682	0.00978	0.01567	0.00972	0	0.06736	0.00919
0.02416	0	0.08333	0.00227	0.09214	0.06093	0.09055	0.02164	0.01295	0.04779
0.02809	0.01011	0.01123	0.00227	0.00747	0.00696	0.01211	0.01716	0.03368	0.03309
0.02345	0	0.05317	0	0.05858	0.02577	0.05812	0.0097	0.01036	0.05699
0.01409	0.08466	0.05802	0.05061	0.05178	0.04492	0.06782	0.03582	0.02073	0.04412
0.06268	0.00126	0.02796	0.00227	0.03392	0.02925	0.03903	0.01791	0.05959	0.01103
0.08122	0.04549	0.13543	0.029	0.13441	0.26323	0.10179	0.30149	0.0544	0.00735
0.04446	0.02401	0.03062	0.02673	0.03131	0.0101	0.03739	0.00597	0.00777	0.04963
0.02509	0	0.00993	0.00057	0.00941	0	0.01134	0.00224	0.00777	0.05147
0.05423	0.00126	0.00268	0.00569	0.00337	0.00836	0.00477	0.00672	0.0829	0.00184
0.01567	0	0.03355	0.00512	0.03583	0.00766	0.03668	0.00672	0.01813	0.05515
0.03848	0	0.01525	0.00227	0.01415	0.00348	0.01581	0.00299	0.01295	0.05331
0.00783	0.03412	0.00433	0.0688	0.00319	0.00174	0.0056	0	0.03627	0.04412
0.04342	0.07961	0.06694	0.08359	0.06171	0.10585	0.05973	0.02537	0	0.05882
0.01558	0.2047	0.0314	0.40202	0.02931	0.04004	0.03088	0.02015	0.01554	0.03493

pop_dens	pop_proj	local_emp	dwell_chn	car_work	pt_work	two_cars	walk_work	low_inc	high_inc	
0.15516	0.04644	0.02369	0.04318	0.02911	0.04129	0.02824	0.02086	0.18606	0.0431	
0.15723	0.032	0.03751	0.03174	0.04414	0.06066	0.04622	0.02422	0.17271	0.02868	
0.07978	0.07887	0.08399	0.05672	0.07452	0.18085	0.04364	0.27041	0.17726	0.02061	
0.07715	0	0.14083	0.01208	0.13498	0.08379	0.04364	0.10104	0.08764	0.12973	
0.13822	0	0.11947	0.02453	0.12678	0.07575	0.13031	0.07867	0.0563	0.14155	
0.00403	0.05943	0.11194	0.05287	0.08555	0.12652	0.08056	0.29482	0.01543	0.17	
0.11924	0.01511	0.19999	0.0294	0.20177	0.1968	0.21416	0.11751	0.1203	0.1083	
0.13189	0.10914	0	0.08446	0	0.03732	0	0.0336	0.22822	0	
0.14602	0	0.15519	0.01087	0.1759	0.10852	0.17455	0.0336	0.08034	0.11278	
0.04513	0	0.04134	0	0.01919	0.02744	0.03931	0	0	0.21157	
0.16569	0	0.16367	0.01384	0.17812	0.13343	0.18293	0.06276	0.1079	0.10372	
0.12904	0.06352	0.0142	0.05481	0.01568	0.04765	0.01909	0.03655	0.22822	0	
0.15116	0	0.15579	0.01938	0.13931	0.11107	0.1602	0.10294	0.04735	0.1598	
0	0.36583	0.10017	0.33166	0.0931	0.00277	0.12117	0.19007	0.10139	0.12973	
0.14109	0.03701	0.03507	0.03403	0.04525	0.06512	0.04506	0	0.18171	0.0431	
0.08996	0	0.20707	0.01384	0.2197	0.17049	0.21749	0.08296	0.0563	0.14533	
0.10035	0.04644	0.0504	0.01384	0.03658	0.03459	0.05344	0.06977	0.1142	0.11278	
0.088	0	0.15601	0	0.16621	0.09427	0.16537	0.04497	0.04735	0.16326	
0.06006	0.20904	0.16518	0.151	0.1533	0.13937	0.18249	0.11926	0.08034	0.13769	
0.17361	0.00843	0.10001	0.01384	0.11478	0.1033	0.12659	0.07204	0.16805	0.04971	
0.20392	0.14057	0.27076	0.10267	0.26974	0.35134	0.23257	0.36149	0.15839	0.03612	
0.13841	0.08953	0.10675	0.0968	0.10845	0.0464	0.12287	0.03057	0.03775	0.14905	
0.09245	0	0.04579	0.00425	0.04392	0	0.05078	0.01366	0.03775	0.1527	
0.15806	0.00843	0.01585	0.0294	0.01919	0.03998	0.02548	0.0336	0.20643	0.01158	
0.06512	0	0.1139	0.027	0.11929	0.03732	0.12124	0.0336	0.07272	0.1598	
0.12535	0	0.06381	0.01384	0.06026	0.01971	0.06557	0.01736	0.0563	0.15628	
0.03796	0.11525	0.02358	0.18415	0.01833	0.01106	0.02905	0	0.1203	0.13769	
0.13619	0.20146	0.18101	0.20746	0.17188	0.23771	0.16832	0.09322	0	0.16666	
0.06483	0.3247	0.10867	0.36634	0.10345	0.12885	0.10739	0.07867	0.06473	0.11716	
0.94292	0.57946	0.88844	0.60108	0.88157	0.8058	0.89025	0.71815	0.89433	0.92018	
0.05708	0.42054	0.11156	0.39892	0.11843	0.1942	0.10975	0.28185	0.10567	0.07982	1.87782
0.0304	**0.22395**	**0.05941**	**0.21244**	**0.06307**	**0.10342**	**0.05844**	**0.15009**	**0.05628**	**0.04251**	

1.5 Council-wide Land Suitability Results

This weighted sum overlay approach acknowledged explicit spatial differences for metro station suitability across Liverpool Council. The highest suitability scores were assigned to suburbs that combine future population growth, dwelling growth, employment activity, public transport demand and others.

Table 5 Council-wide Land Suitability Scores

Rank	Suburb	Suitability Score	Key reasons
1	Liverpool	0.541	Highest density, strongest local employment, most number of people using car to work alongside public transport ridership and walking count
2	Badgerys Creek	0.465	High population projection and dwelling growth because of its relevance to airport and aerotropolis
3	Austral	0.435	Highest forecast dwelling change and very high population projection
4	Edmondson Park	0.292	Strong public transport usage, high employment/labour force, high dwelling growth and existing rail connectivity
5	Moorebank	0.245	Strong population projection, employment base and households with two or more cars
6	Casula	0.231	High employment/labour force, people using car to work and households with tow or more cars
7	Prestons	0.218	High local employment, number of people using car to work and households with two or more cars.
8	Holsworthy	0.191	High number of people walking to work and strong public transport connectivity

9	Warwick Farm	0.183	High public transport and number of people walking to work alongside low-income household concentration
10	Green Valley	0.180	High population density and number of people using car to work. High number of households with two or more cars.

The highest ranked suburb was Liverpool, with a suitability score of 0.541. Liverpool has recorded the highest population density (5,518 persons per sqkm), the highest local employment/labor force count (11,241), the highest number of people travelling to work by car (4,727), the highest number of people using public transportation to work (771) and those who walk to work (404). This validates Liverpool City Centre as the strongest node.

Followed by Badgerys Creek with a score of 0.465. It is driven by high population projection value of +322.4% and forecasted dwelling change of +383%. Austral ranked third with 0.435, supported by highest forecasted dwelling change of all at +708% and population projection value of +162%. This implies that western growth corridor is critical for future station planning too.

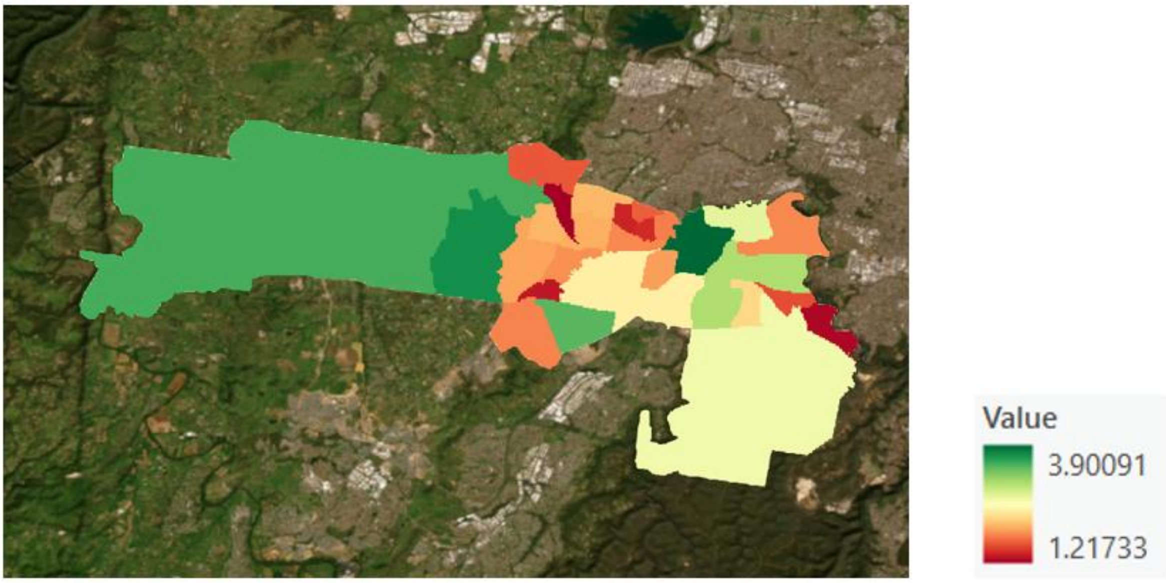

Figure 2 Liverpool Council Land Suitability Map (made by author on MS Powerpoint)

1.6 Transport Dependency on Growth Pressure and Future Demand

This suburb-level analysis displays robust variance in future growth, transport dependency and commuting behaviour across Liverpool Council. Current high-density suburbs like Liverpool, Green Valley, and Sadleir contain some of the largest residential concentrations, while western growth areas like Badgerys Creek, Austral, and Edmondson Park show strongest projected population and dwelling growth to 2046. Wherein, Austral observed the highest forecasted dwelling change at around +708%, while Badgerys Creek showed highest forecasted population projection value within the dataset. These western and south-western growth areas are thus expected to generate substantial future transport demand associated with the Aerotropolis and Western Sydney International Airport. Hence, metro station network should realise both existing demand in Liverpool City Centre and future demand in the western growth areas in placement of their stations.

Table 6 Transport Dependency on Growth Pressure and Future Demand

Indicator	Highest suburbs	Interpretation for station demand
Population projection to 2046	Badgerys Creek (322.4), Austral (162), Moorebank (67), Edmondson Park (63), Liverpool (36)	Future population growth is strongest in airport/Aerotropolis and western growth areas.
Forecast dwelling change to 2046	Austral (708%), Badgerys Creek (383%), Edmondson Park (148%), Denham Court (122%), Moorebank (90%)	Strong dwelling growth suggests future TOD and new commuter demand.
Existing population density	Liverpool (5,518), Lurnea (4,272), Green Valley (3,977), Sadleir (3,704), Busby (3,675)	Existing density is strongest in established suburbs, but not always aligned with future growth.

These transport indicators also unveil strong car dependency across most of Liverpool Council. Suburbs like Prestons, Casula, Green Valley, and Hinchbrook recorded high number of people using car to work alongside households with two or more cares, implying their heavy reliance on private vehicles for daily commuting. Liverpool itself recorded the highest car-to-work value at 4,727 commuters despite having highest public transport-to-work count at 771 commuters. Walking to work is also concentrated in specific location like Liverpool, Holsworthy, and Warwick Farm. This suggests station suitability should also include land use mix and employment proximity. Thereby, a future metro station should either target highly car-dependent areas where mode shift is possible or those areas where existing public transport ridership is high.

Table 7 Mode of Travel to Work and Car dependency

Indicator	Highest suburbs	Interpretation
Car travel to work	Liverpool (4,727), Prestons (3,335), Casula (2,929), Green Valley (2,450), Hinchinbrook (2,408)	Existing commuter demand is highly car-oriented.
Public transport to work	Liverpool (771), Edmondson Park (319), Casula (235), Warwick Farm (207), Prestons (190)	Existing PT demand is strongest near current rail/centre-based locations.
Two or more car households	Liverpool (3,452), Prestons (3,103), Casula (3,030), Green Valley (2,405), Moorebank (2,397)	Car ownership is embedded in both established suburbs and employment corridors.
Walking to work	Liverpool (404), Holsworthy (217), Warwick Farm (181), Badgerys Creek (97), Moorebank (48)	Walking is strongest where jobs and housing are closer or where institutional/employment land exists.

Moving to relationship between income and transport accessibility, it suggested possible transport disadvantage in suburbs like Cartwright, Miller, Sadleir, Heckenberg and Busby as they notice high reliance on private vehicles despite falling under the bracket of lower income. This goes against suburbs like Liverpool,and Warwick farm, which demonstrated higher usage of public transport along with a greater number of high-income households. An influential reason can be the presence of existing rail accessibility, employment concentration and proximity to major centres. Therefore, equity-based station citing should not only consider areas with low-income but also ones that overlap this with low public transport usage and higher car dependency.

On the whole, these combined results suggest that future metro planning should balance both existing commuter demand and future population growth for sustainable outcomes. They form basis for identifying more appropriate areas for construction of metro station in the next stage of spatial visualization and site screening.

1.7 Council-wide suitability Map suggestions for Preliminary Site Selection

This land suitability map and scoring put forward a wider spatial evidence base for shortlisting potential metro station areas. Based on the weighted overlay results and differences between scores obtained, the strongest preliminary opportunity areas are:

Table 8 Summarization of Council-wide assessment

Preliminary Opportunity Area	Evidence-based Reasoning	Planning Interpretation
Liverpool City Centre	Highest suitability score (0.541), strongest employment, public transport ridership and walking values	Strong existing demand and interchange potential
Badgerys Creek (Aerotropolis Interface)	Second-highest suitability score (0.465), strongest population projection value (322.4)	Very strong strategic value in the future growth scenario.
Austral	Third highest suitability score (0.435), highest dwelling growth forecast (708%) and strong future population growth.	Strong long-term TOD/growth corridor case.

2. Detailed Land Suitability Analysis and Site Narrowing

2.1 Transition from Council-level Screening to Detailed Spatial Assessment

Previously covered suburb-level indicators like population growth, car dependency, public transport usage, percentage of dwelling growth, and household income level to conduct a council-wide land suitability assessment using. That phase was intended to recognize broad strategic opportunity areas across Liverpool Council rather than determining exact locations for the construction of metro station.

The weighted overlay results identified Liverpool, Badgerys Creek, and Austral as three strongest opportunity areas within the council boundary. However, suburb-based indicators alone are not sufficient for determining precise metro station locations. For instance, high public transport usage within a suburb does not necessarily indicate that all land parcels within that suburb are appropriate for building a metro station.

Therefore, this phase narrows down the analysis from the council scale to a more detailed precinct-scale spatial assessment. Herein, additional layers linked with environmental, heritage, zoning and transport infrastructure were incorporated to identify more realistic station opportunity zones.

2.2 Detailed Land Suitability Analysis – Liverpool LGA

Liverpool witnessed the highest overall suitability score in the council-wide assessment and therefore became the priority in the detailed study area. Unlike the previous council-level analysis, the Liverpool LGA uses more targeted demographic and travel behaviour indicators derived directly from ABS TableBuilder datasets.

This detailed demographic analysis exposed that public transport dependency within the Liverpool LGA is not evenly distributed among all population groups. In fact, the younger population, university-educated residents, part-time employees, and low-income residents tend to use public transportation the most. It is true that number of people relying on car for daily commute is considerable among low-income households as well. However, the percentage share of residents using public transportation is relatively higher than other income brackets as seen in the forthcoming paragraphs.

Table 9 Age and Method of Travel - Liverpool LGA

	Liverpool (15-19)	Liverpool (20-24)	Liverpool (25-29)	Liverpool (30-34)	Liverpool (35-39)	Liverpool (40-44)	Liverpool (45-49)	Liverpool (50-54)	Liverpool (55-59)	Liverpool (60-64)	Liverpool (65-69)	Liverpool (70-74)	Liverpool (75-79)
MTW15P Method of Travel to													
Train	88	309	269	276	259	201	142	149	124	103	35	11	0
Bus	29	59	79	35	33	39	35	43	31	29	11	0	0
Taxi/ride-share service	8	17	17	12	12	6	4	11	5	7	0	0	0
Car, as driver	608	2139	2426	2532	2650	2581	2573	2396	1998	1431	541	149	38
Car, as passenger	511	259	164	146	90	102	119	113	106	89	35	7	4
Truck	0	9	34	41	36	42	48	30	31	20	8	6	0
Motorbike/scooter	3	16	26	16	11	9	12	11	12	6	0	0	0
Bicycle	4	24	20	21	10	7	12	4	9	4	0	0	0
Other Mode	16	41	30	36	43	40	41	31	27	10	6	0	0
Walked only	145	179	162	123	87	97	91	74	72	51	21	8	0
Worked at home	145	1262	2096	2297	2671	2275	1966	1499	1083	654	244	96	29
Did not go to work	837	1098	906	1113	1041	778	720	693	628	435	234	70	29
Not stated	9	30	29	40	26	28	25	21	23	15	11	9	4
Not applicable	6328	3965	3724	3978	3857	3452	3125	2993	3357	3682	3905	3774	2868
Total	8735	9411	9995	10646	10816	9651	8900	8081	7504	6533	5052	4132	2972

The younger age groups, especially 15-24 years display relatively stronger public transport dependency in comparison with middle-aged population groups, who are more likely to drive to work.

Table 10 Income and Method of Travel - Liverpool LGA

	Liverpool (500-649)	Liverpool (650-799)	Liverpool (800-999)	Liverpool (1000-1249)	Liverpool (1250-1499)	Liverpool (1500-1749)	Liverpool (1750-1999)	Liverpool (2000-2999)	Liverpool (3000-3499)	Liverpool (3500 and more)
MTW15P Method of Travel to										
Motorbike/scooter	5	7	20	27	17	11	120	12	3	4
Train	162	244	317	360	220	123	1962	155	21	22
Bus	47	62	86	57	39	20	422	3	0	0
Taxi/ride-share service	12	12	18	21	10	3	95	3	0	0
Car, as driver	1332	2006	3353	4099	2991	2224	22066	1882	303	353
Car, as passenger	141	203	282	217	113	59	1746	46	0	11
Bicycle	7	24	14	20	9	13	122	11	0	0
Other Mode	19	39	36	48	43	35	314	29	4	3
Walked only	81	111	182	207	115	99	1102	55	7	11
Worked at home	418	639	1183	2101	2243	2184	16323	3453	692	784
Not stated	19	32	37	30	19	15	273	20	0	3
Not applicable	4038	2532	1862	1367	794	560	82710	303	59	103
Did not go to work	741	1043	1109	1198	853	680	8602	586	86	57
Truck	16	30	39	68	38	28	304	30	3	6
Total	7047	6993	8534	9825	7503	6056	136165	6588	1188	1356

Lower income groups hold higher share of public transport users of all. As they income increases, the share of public transportation ridership continues to fall.

Table 11 Education and Employment with Method of Travel

	Liverpool	Total	Secondary Education (Years 10 and above)	Bachelor's Degree	Postgraduate Degree	Full-time Employees	Part-time Employees
MTW15P Method of Travel to							
Motorbike/scooter	120	120	31	16	7	90	33
Train	1962	1962	533	552	222	1146	594
Bus	422	422	157	80	31	186	198
Taxi/ride-share service	95	95	47	19	5	52	40
Car, as driver	22066	22066	6786	4267	1468	13566	6759
Car, as passenger	1746	1746	885	222	80	719	884
Bicycle	122	122	43	12	7	72	34
Other Mode	314	314	103	57	21	160	86
Walked only	1102	1102	403	209	99	659	381
Worked at home	16323	16323	2832	5837	2497	12479	3384
Not stated	273	273	85	38	19	40	26
Not applicable	82710	82710	18957	4097	1162	0	0
Did not go to work	8602	8602	3112	1353	394	1041	1908
Truck	304	304	126	16	0	166	107
Total	136165	136165	34107	16781	6013	30379	14427

Residents with undergraduate and postgraduate qualifications display higher acceptance rates towards riding a public transport facility. Part-time workers recorded a strong usage of public transportation in comparison to full-time employees.

These insights specifically identify socially and behaviourally transport dependent groups within Liverpool LGA. Thereafter, to ensure objective weighting of indicators, Shannon entropy weighting was again used because the Liverpool assessment includes comparison between multiple demographic variables and travel attributes across the LGA. Entropy weighting permits indicators with stronger variation and discriminatory power to receive higher influence in the final suitability model.

Table 12 Normalised, Proportionate, and Entropy-derived tabular values of the indicators - Liverpool LGA

fif_twe	less_inc	ug_pg	rtime_em	er_ar_pe	pop_dens	pt_stops
0.41	0.09	0.65	0.40	1.00	0.23	0.09
0.87	0.60	0.46	0.62	1.00	0.19	0.32
0.74	0.84	0.48	0.62	0.98	0.28	0.45
0.48	0.67	0.00	0.17	1.00	0.45	1.00
0.50	0.81	0.17	0.29	0.96	0.41	0.91
0.64	0.92	0.71	0.58	0.67	1.00	0.59
0.00	0.00	0.02	0.00	0.74	0.13	0.14
0.78	1.00	0.75	0.93	0.94	0.15	0.23
1.00	0.76	1.00	1.00	0.00	0.00	0.00
5.43	5.70	4.24	4.60	7.29	2.85	3.73

fif_twe	less_inc	ug_pg	rtime_em	er_ar_pe	pop_dens	pt_stops
0.07	0.02	0.15	0.09	0.14	0.08	0.02
0.16	0.11	0.11	0.13	0.14	0.07	0.09
0.14	0.15	0.11	0.13	0.13	0.10	0.12
0.09	0.12	0.00	0.04	0.14	0.16	0.27
0.09	0.14	0.04	0.06	0.13	0.15	0.24
0.12	0.16	0.17	0.13	0.09	0.35	0.16
0.00	0.00	0.00	0.00	0.10	0.05	0.04
0.14	0.18	0.18	0.20	0.13	0.05	0.06
0.18	0.13	0.24	0.22	0.00	0.00	0.00

fif_twe	less_inc	ug_pg	rtime_em	er_ar_pe	pop_dens	pt_stops
0.19	0.07	0.29	0.21	0.00	0.20	0.09
0.29	0.24	0.24	0.27	0.00	0.18	0.21
0.27	0.28	0.25	0.27	0.27	0.23	0.26
0.22	0.25	0.00	0.12	0.00	0.29	0.35
0.22	0.28	0.13	0.17	0.27	0.28	0.34
0.25	0.29	0.30	0.26	0.22	0.37	0.29
0.00	0.00	0.02	0.00	0.23	0.14	0.12
0.28	0.31	0.31	0.32	0.26	0.16	0.17
0.31	0.27	0.34	0.33	0.00	0.00	0.00
0.93	0.90	0.85	0.89	0.57	0.84	0.84
0.07	0.10	0.15	0.11	0.43	0.16	0.16
0.061657	**0.082241**	**0.125922**	**0.090907**	**0.365651**	**0.134534**	**0.139088**

The weighted overlay performed in ArcGIS Pro generated a highly curated land suitability raster surface where:

- Areas in green present highly favourable zones for metro station
- Areas in yellow/orange depict moderate suitability
- Areas in red warn about constrained locations

As can be seen in below, regions near existing transport corridors, higher public transport activity and stronger mixed urban activity witnessed higher suitability while environmentally constrained and low-accessibility areas were regarded as relatively less significance.

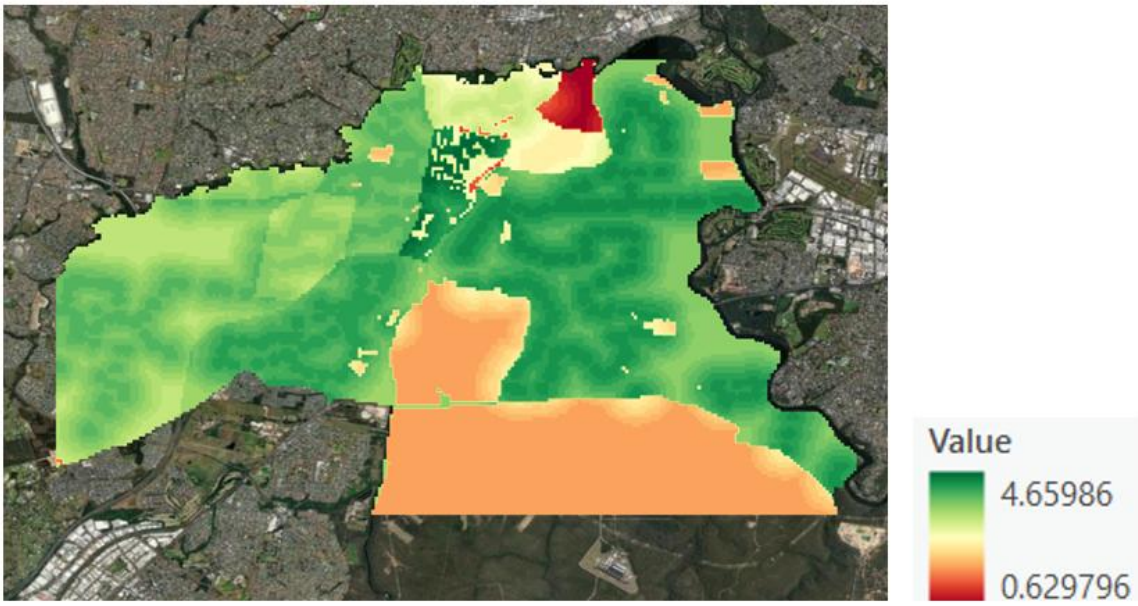

Figure 3 Land Suitability Analysis - Liverpool LGA (made by author on MS Powerpoint)

2.3 Detailed Land Suitability Analysis – Badgerys Creek and Aerotropolis Interface

Badgerys Creek obtained the second highest suitability score in the council-wide assessment and was considered strategically important because of its direct relationship with:

- Western Sydney International Airport
- The Aerotropolis
- Major State-led Infrastructure Projects

Unlike Liverpool, the Badgerys Creek assessment relied less on demographic comparison and more on strategic spatial planning regulations derived from State Environmental Planning Policy (SEPP) mapping layers available through Liverpool Council's Planning Portal. These included:

- Aerotropolis boundary mapping
- Heritage mapping
- Land Application mapping
- Land Zoning mapping
- Major Transport Corridor mapping

This Major Transport Corridor map recognized numerous strategic elements like:

the proposed metro alignment

- East-West Rail Link
- M12 Motorway
- Elizabeth Drive
- Fifteenth Avenue Corridor
- Outer Sydney Orbital Corridor

Similarly, the land zoning map highlighted the spatial distribution of:

- Environmental and Recreation Lands
- SP2 Infrastructure Corridors.

Heritage and environmental layers were particularly important because vast parts of the Aerotropolis precinct contain environmentally sensitive and preserved areas that cannot be disturbed for the purpose of metro station construction.

Since this stage focused on a single strategic precinct rather than a comparison across multiple spatial units, Analytical Hierarchy Process (AHP) weighting was used instead of entropy weighting. This way of direction was more appropriate because:

1. The data structure involved spatial planning criteria rather than large statistical comparisons
2. And pairwise judgement between planning indicators was required.

Herein, pairwise comparison matrices were developed using, permitting indicators revolving around transport integration, environmental protection, and land use compatibility to be prioritised accordingly. And the final raster-based suitability map was again classified as:

- Green zones as highly suitable.
- Yellow/Orange zones as moderately suitable
- Red zones as unsuitable

As can be seen below, the strongest favourable zones generally align with future transport corridors located outside of major environmental or heritage constraints.

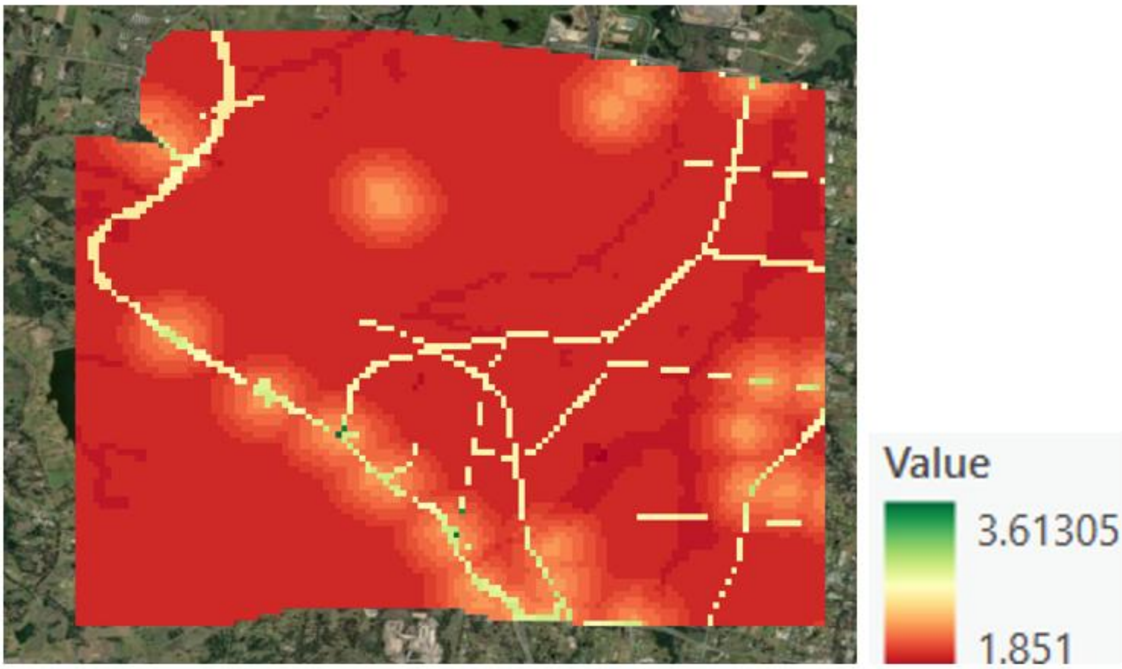

Figure 4 Land Suitability Analysis - Western Parkland City, Badgerys Creek (made by author on MS Powerpoint)

2.4 Detailed Land Suitability Analysis - Austral

Austral ranked as third most suitable area in the council-wide assessment due to its strong performance in future dwelling and population growth projections. While Liverpool represented existing urban demand and Badgerys Creek uplifted airport-oriented strategic growth, Austral embodied long-term residential expansion. Herein, a similar methodology to that of Badgerys Creek assessment with exclusive emphasis on:

- Heritage preservation areas
- Exact Public Transport Stop locations extracted from OpenStreetMap (OSM)

Unlike the council-level assessment, which only considered number of public transport users within a suburb, this stage analyses the precise spatial location of the public transport infrastructure. This permits the study to realise:

- Transport gaps
- Areas with weaker existing accessibility
- And locations where proposed metro station can improve the spatial integration

AHP weighting was used again for the same logic as before, herein assessment acknowledged spatial planning constraints and transport accessibility within a single growth precinct. And the generated raster-based land suitability map displays clear distinction between highly accessible growth-oriented lands and those hindered with heritage (incompatible land use). Evidently, suitable zones in green tend to concentrate near transport corridors and developing residential areas while the red zone (unsuitable) are mostly surrounded by environmentally sensitive land alongside locations with weaker accessibility.

Table 13 Pairwise Matrix Comparison - Austral

	pop_dens	pop_proj	local_emp	dwell_chng	car_work	pt_work	two_cars	walk_work	low_inc	high_inc	her_ar_per	pts_sqkm	
pop_dens	0.088496	0.130719	0.122699	0.086022	0.09009	0.077008	0.075282	0.092879	0.098765	0.098765	0.090909	0.076726	**0.09403**
pop_proj	0.044248	0.065359	0.06135	0.086022	0.09009	0.046205	0.062735	0.092879	0.098765	0.098765	0.090909	0.046036	**0.073614**
local_emp	0.044248	0.065359	0.06135	0.086022	0.09009	0.046205	0.062735	0.092879	0.098765	0.098765	0.090909	0.046036	**0.073614**
dwell_chng	0.044248	0.03268	0.030675	0.043011	0.06006	0.038504	0.050188	0.06192	0.049383	0.049383	0.072727	0.038363	**0.047595**
car_work	0.026549	0.019608	0.018405	0.021505	0.03003	0.030803	0.040151	0.03096	0.024691	0.024691	0.054545	0.030691	**0.029386**
pt_work	0.176991	0.196078	0.184049	0.172043	0.15015	0.154015	0.125471	0.154799	0.148148	0.148148	0.127273	0.153453	**0.157552**
two_cars	0.265487	0.261438	0.245399	0.215054	0.18018	0.308031	0.250941	0.154799	0.197531	0.197531	0.145455	0.306905	**0.227396**
walk_work	0.026549	0.019608	0.018405	0.021505	0.03003	0.030803	0.050188	0.03096	0.024691	0.024691	0.054545	0.030691	**0.030222**
low_inc	0.044248	0.03268	0.030675	0.043011	0.06006	0.046205	0.062735	0.06192	0.049383	0.049383	0.072727	0.046036	**0.049922**
high_inc	0.044248	0.03268	0.030675	0.043011	0.06006	0.046205	0.062735	0.06192	0.049383	0.049383	0.072727	0.046036	**0.049922**
her_ar_per	0.017699	0.013072	0.01227	0.010753	0.009009	0.022002	0.031368	0.009288	0.012346	0.012346	0.018182	0.025575	**0.016159**
pts_sqkm	0.176991	0.130719	0.184049	0.172043	0.15015	0.154015	0.125471	0.154799	0.148148	0.148148	0.109091	0.153453	**0.15059**

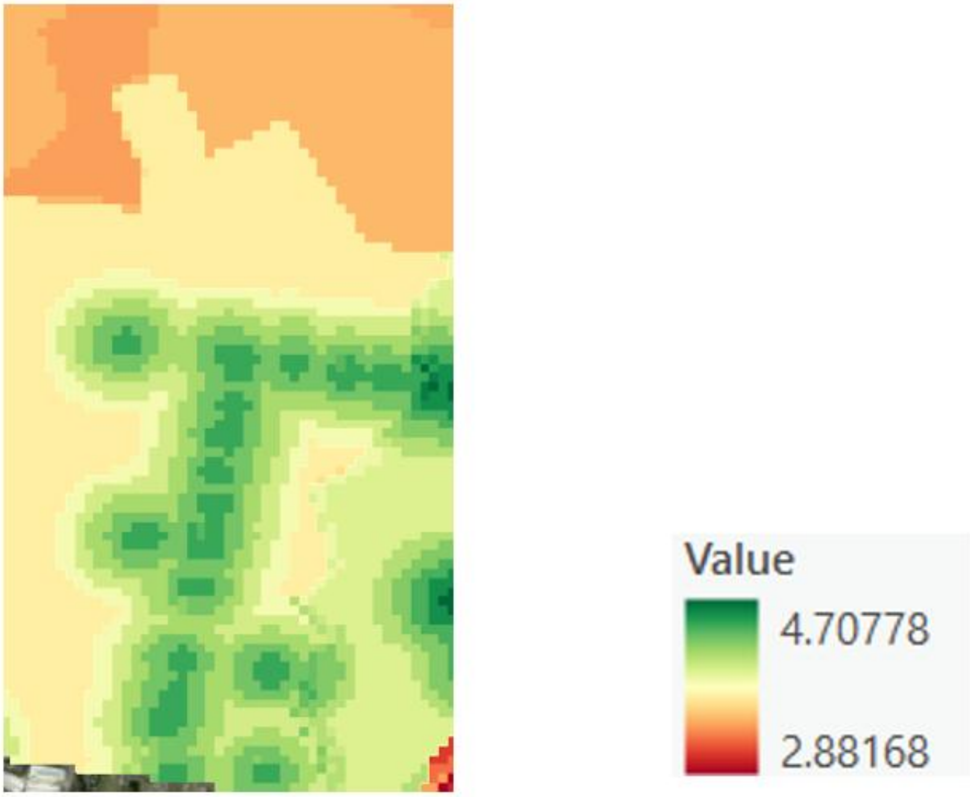

Figure 5 Land Suitability Analysis – Austral (made by author on MS Powerpoint)

2.5 Role of Kepler.gl in Spatial Visualisation

Following the GIS-based land suitability analysis, the outputs were visualised in kepler.gl to enhance spatial interpretation and comparative understanding of the study areas. While ArcGIS Pro was primarily used for raster generation, weighted overlay analysis and spatial preprocessing, Kepler.gl enabled interactive visualisation of various demographic, transport and other planning indicators within a three-dimensional environment.

Figure 6 Kepler Visualisation of Public transport users (3D columns) and Local Employment Force (Flat surface) (made by author on MS Powerpoint)

The first Kepler visualisation highlighted the broader Liverpool Council geography and comined:

- Number of people using public transport
- Local employment concentration (suburb-wise)

As can be seen in (), flat-surfaced aggregation layers are used to portray local employment concentration across suburbs while 3D extruded columns represent public transport ridership.

The combination of these layers revealed a clear spatial relationship between:

- Higher employment concentration
- Stronger public transport usage
- Existing urban centres

Liverpool LGA demonstrated the strongest clustering of employment activity and transport usage, strengthening its high suitability score obtained through entropy-based weighted overlay analysis. This visualisation also highlighted the contrast between established urban centres and emerging western growth areas.

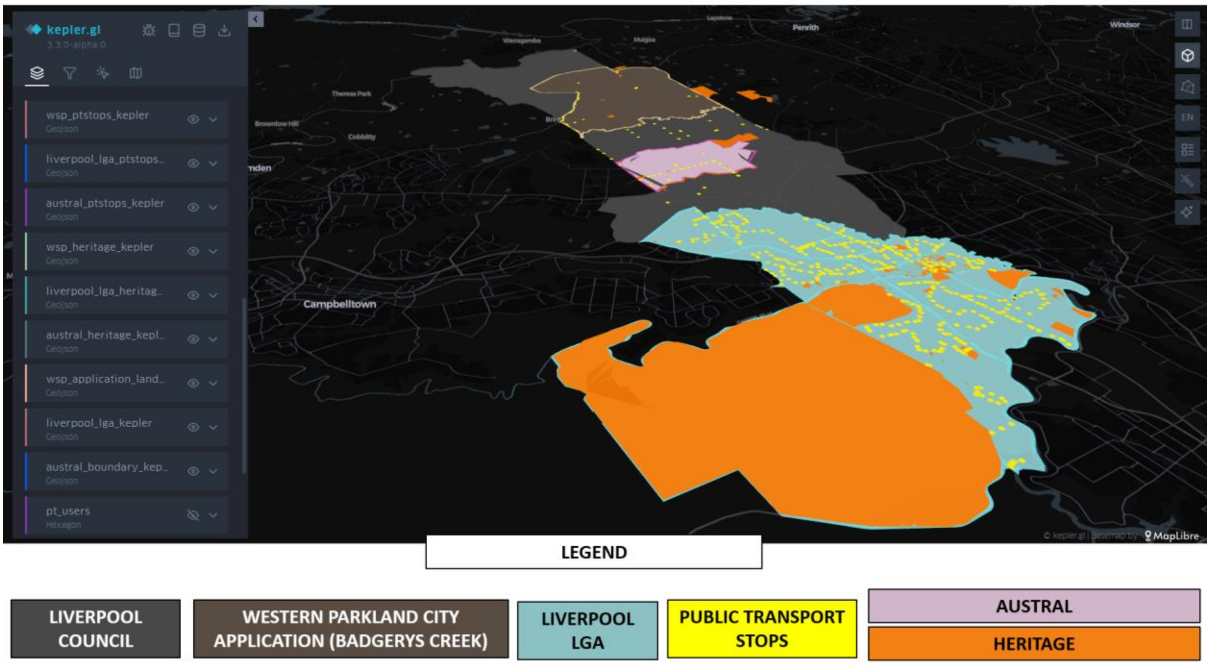

Figure 7 Kepler visualisation of heritage and public transport stop layers in the favourable areas of Council-level Land Suitability Analysis (made by author on MS Powerpoint)

Herein, the second kepler visualisation integrated:

- Liverpool Council boundaries
- Liverpool LGA boudnaries
- Austral suburb boundary
- Western Parkland City application areas within Badgerys Creek
- Public transport stop locations and heritage layers (extracted from SEPP) within all of these.

This visualisation is particularly important because it depicted how accessibility and environmental constraints influenced the overall performance of these areas in their enumerations. Unlike static GIS outputs, Kepler,gl allowed multiple datsets to be visualised together dynamically, enhancing study of:

- Spatial clustering
- Transportation concentration
- Planning constraints

This platform was particularly helpful in communicating the broader strategic geography of the study in a visually intuitive way.

3. Walkability Catchment Assessment and Site Filtering

3.1 Transition from Land Suitability to Walkability Assessment

Previously, suitable opportunity zones for constructing metro station have been realised using demographic, environmental, transport, and other planning -related indicators in raster-based land suitability analysis. However, that does not alone guarantee that a station location can effectively serve its commuters. A metro station must also be accessible within a comfortable walking distance to maximize ridership and shift dependency from private vehicles.

Subsequently, succeeding stage of this analysis focuses on pedestrian accessibility and catchment performance. The point locations of these sites were selected from the darkest green zones within the raster output of the land suitability analysis as they represent the conditions for most favourable areas for this context.

These selected points were then tested on PedestrianCatch.com, which simulates walkable catchments based on actual street-network movement rather than conventional circular buffers. The assessment was performed under:

- An average walking speed of 1.33 m/s, representing average adult walking speed,
- And a 10-minute walking duration, aligning with the approximate desirable walking distance promoted within NSW transit-oriented development principles and station catchment planning.

Thereafter, these generated catchments were further analysed using the Australian Urban Observatory (AUO). Herein, selected station points are placed as centre in the map and the tool estimates the catchment for the selected duration, 10 minutes in our case. Furthermore, it provides us with attributes of the catchment like total population and average household size, giving us the freedom to calculate dwelling units in the catchment.

$$Dwelling\ Units = \frac{Total\ Population}{Average\ Household\ Size}$$

This procedure enabled us to compare between each point locations for the metro construction on basis of number of dwelling units they are serving in a contemporary period within as realistic conditions as possible.

3.2 Liverpool LGA Walkability Assessment

Three possible locations were tested within the Liverpool LGA as per the brief and its stand as the most suitable region within the council-wide assessment.

Figure 8 Potential Sites of Liverpool LGA (Made by author from ArcGIS Pro and MS Powerpoint)

3.2.1 Site A – Hume Highway / Southwestern Motorway

This site is present at the intersection of the Hume Highway and the Southwestern Motorway, slightly south-west of the current Liverpool train station. Although it is not directly adjacent to the current station, its pedestrian catchment successfully connects larger proportions of both Liverpool (northward) and Casula (southward) neighbourhoods. Thereby, creating a potential for the metro station to serve as broader cross-corridor connector rather than encouraging the existing Liverpool Station node.

Figure 9 Site A - Hume Highway / Southwestern Highway (made by author on MS Powerpoint)

The relatively high dwelling coverage interprets strong residential accessibility and strengthens its function as a secondary metropolitan access point beyond the contemporary Liverpool train station.

3.2.2 Site B – Newbridge Road / Georges River

Positioned along the Newbridge Road, on the opposite side of the Goerges River in regard to the existing Liverpool train station sits closer to the railway corridor than the previous one. This was selected with a strategic intent of distributing the transit pressure effectively rather than concentrating entirely on one station alone. Following a similar decentralised-access logic to that of approach adopted in Parramatta.

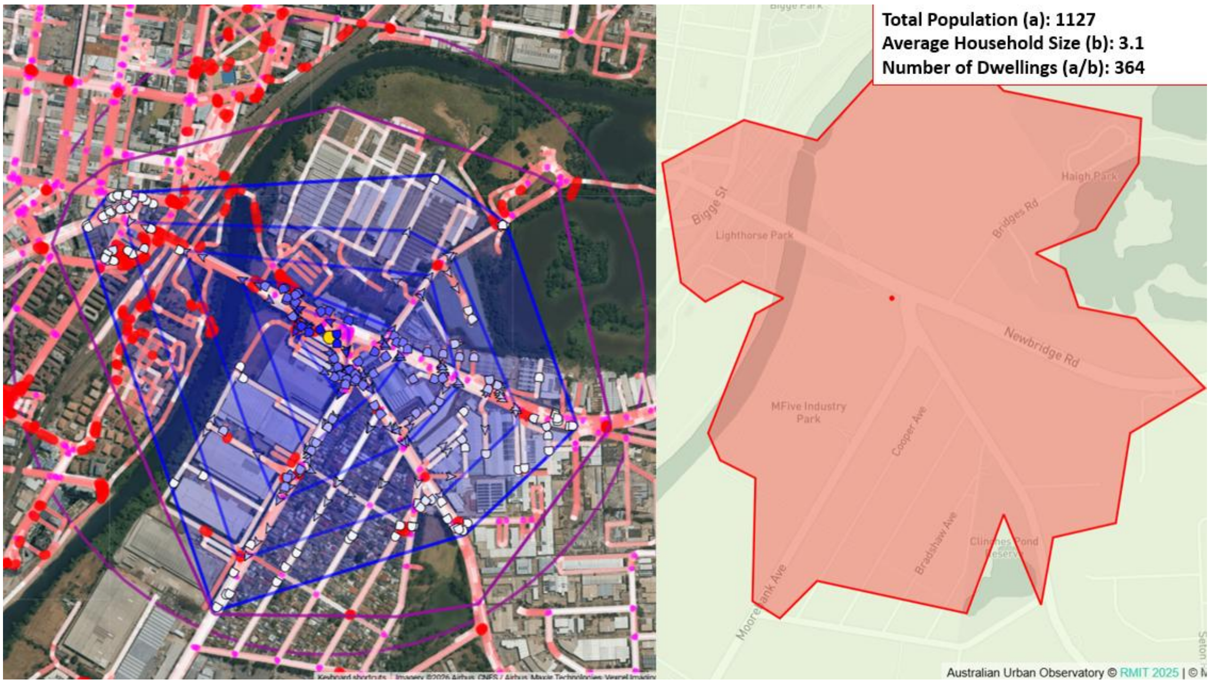

Figure 10 Site B - Newbridge Road / Georges River (made by author on MS Powerpoint)

Although this site recorded lower dwelling coverage in comparison to others, its focus relies on stronger network distribution potential for easing future scenarios.

3.2.3 Site C – A34 / Macquarie Street

Located at the intersection of A34 and Macquarie Street within the broader Liverpool urban core suggests strongest residential catchment among all the potential sites in the LGA. This suggests strongest integration with existing urban grid and surrounding density, leading to immediate ridership potential.

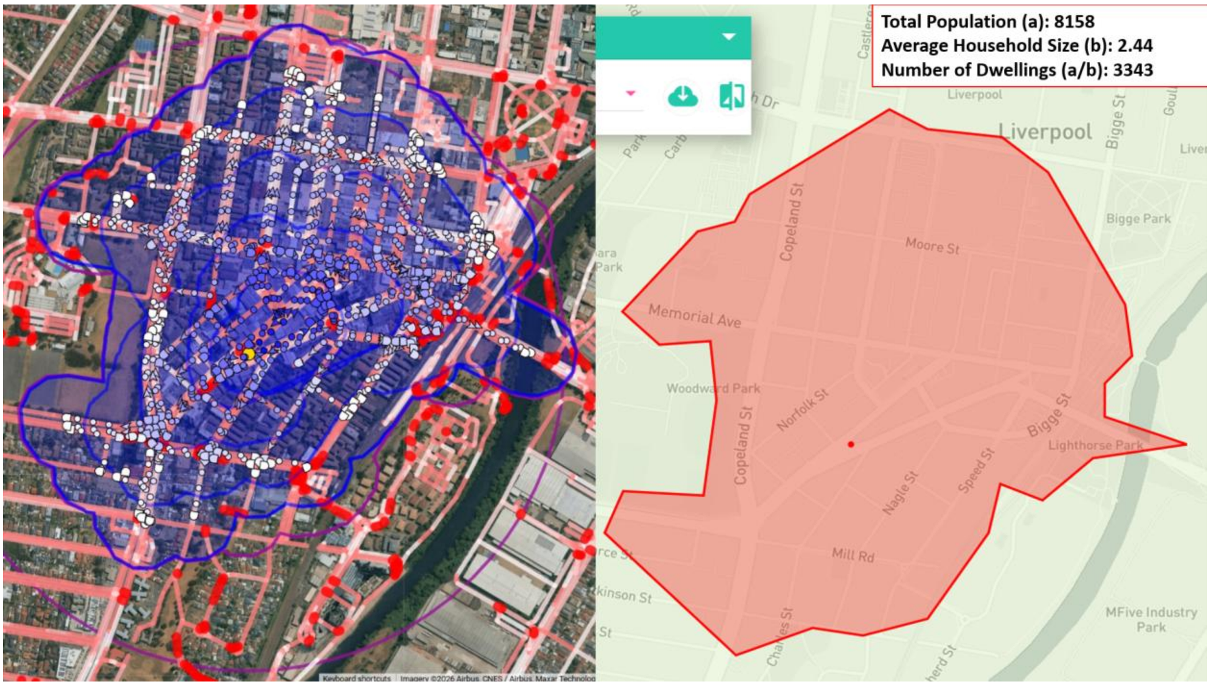

Figure 11 Site C - A34/ Macquarie Street (made by author on MS Powerpoint)

Among the three Liverpool site locations, Site C displays the most promising walkability-based residential catchment performance, followed by Site A. Where, Site B remains strategically important for network distribution, its accessibility performance in present day scenario is quite low.

3.3 Badgerys Creek Walkability Assessment

Two potential sites were tested within this precinct because of its strategic relationship with the Western Sydney International Airport and the Aerotropolis. Unlike Liverpool, the Badgerys Creek is currently a low-density region with partial development. Therefore, its catchment analysis reflects upon contemporary conditions rather than projected scenarios for a fairer comparison.

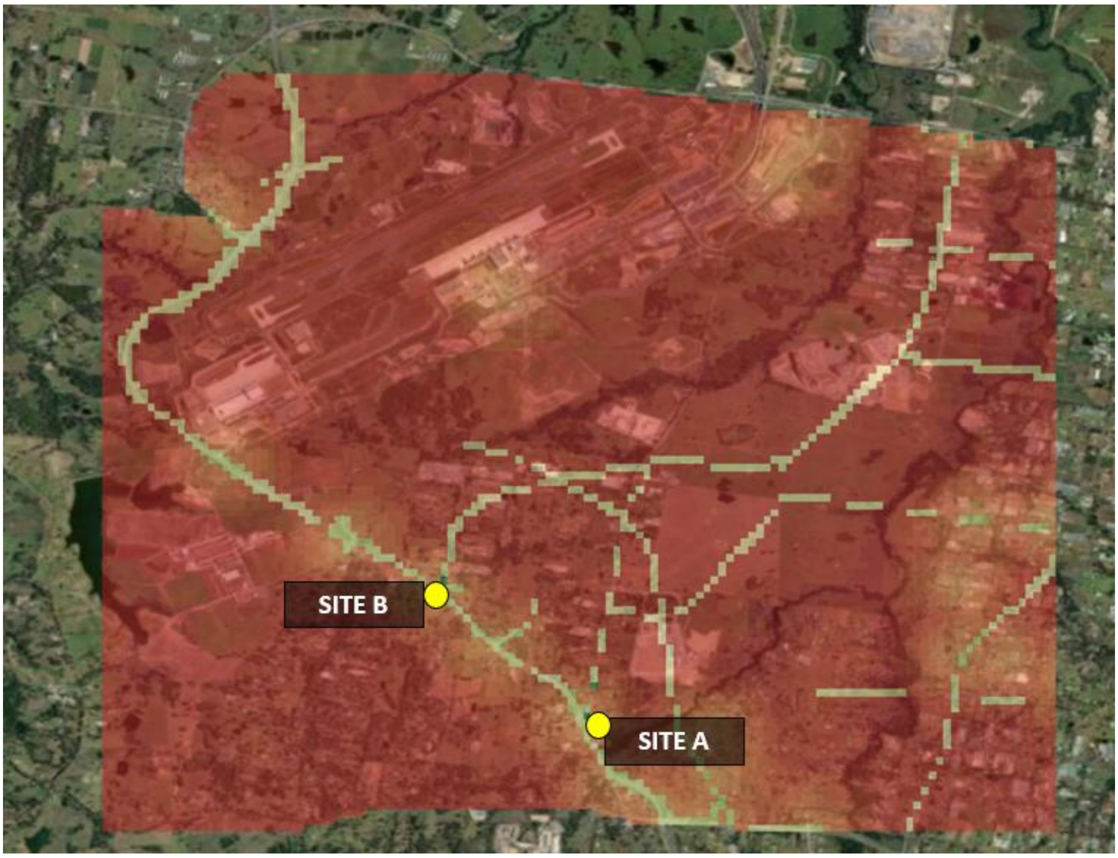

Figure 12 Potential Sites in the Western Parkland Application area (Badgerys Creek) (made by author on MS Powerpoint)

3.3.1 Site A – A9 / Badgerys Creek Road

This is situated at the intersection of A9 and Badgerys Creek Road, whose catchment covers an approximate of 42 dwellings. Although its current dwelling coverage is low, the location remains strategically pivotal because of its placement between both the aerotropolis and the upcoming international airport.

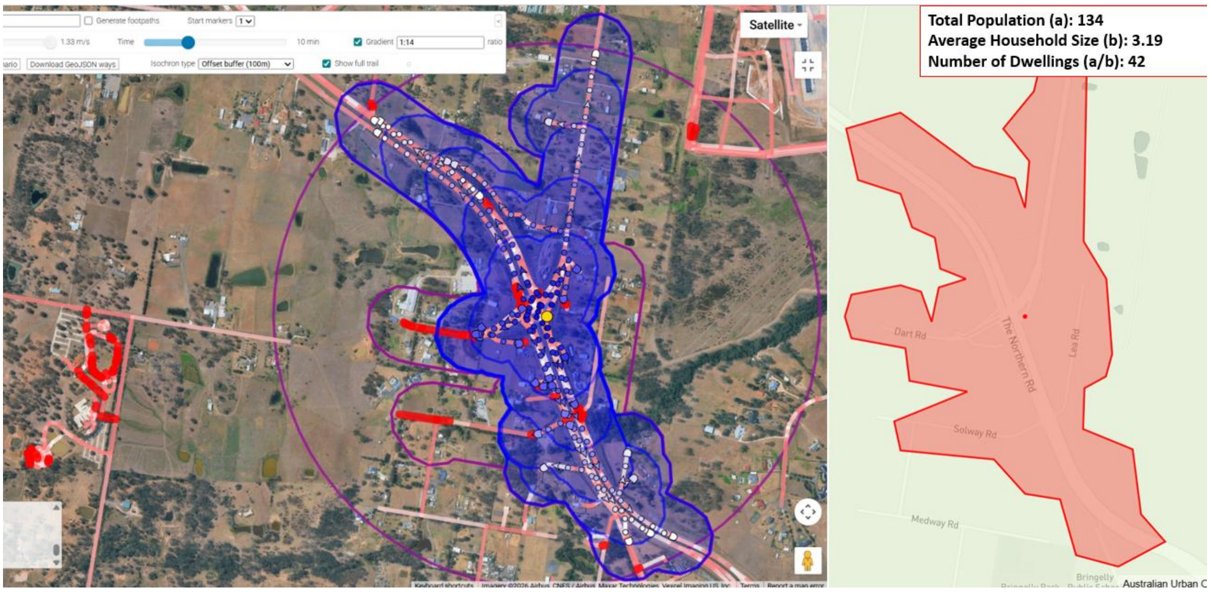

Figure 13 Site A - A9 / Badgerys Creek Road (made by author on MS Powerpoint)

3.3.2 Site B – A9 / Mercy Road

This is presumably placed at the intersection between Mercy Road and A9, positioned closer to the future airport than the previous site. However, its catchment covers even less number of dwelling units than before choice.

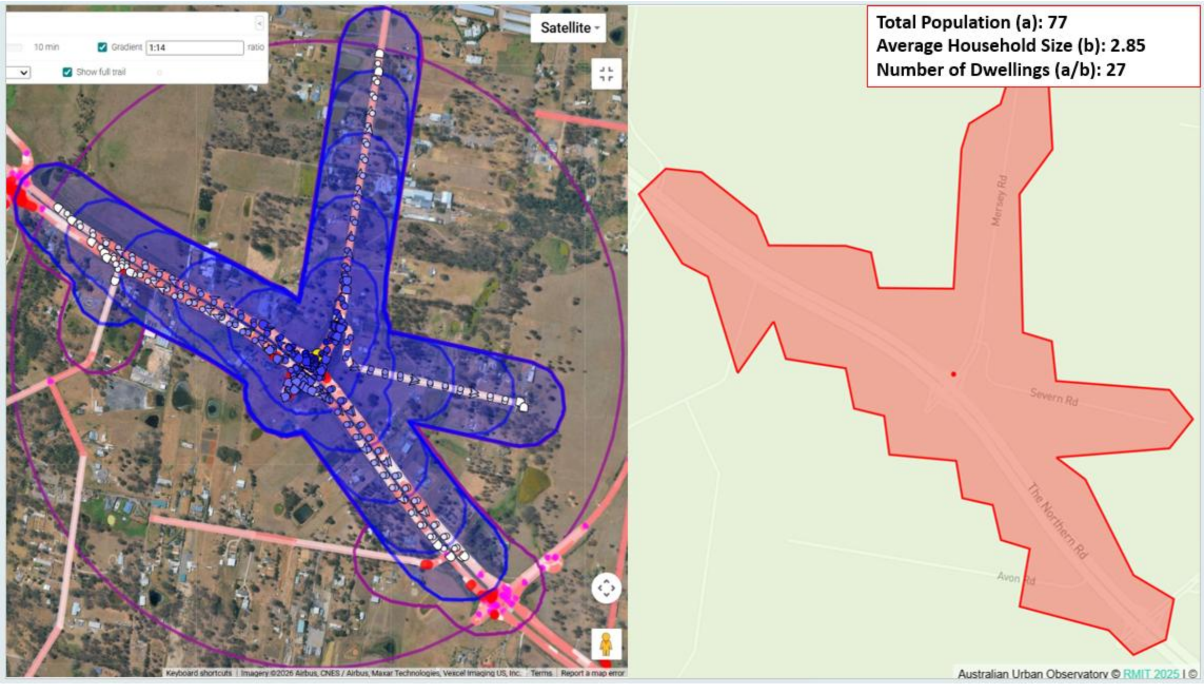

Figure 14 Site B - A9 / Mercy Road (made by author on MS Powerpoint)

Site A of the Badgerys Creek performs better in terms of accessibility than Site B. However, both sites remain strategically future-oriented than immediate demand-oriented.

3.4 Austral Walkability Assessment

On basis of the raster output generated in the land suitability assessment, two potential sites have been chosen for test. Austral had primarily put forward its site on basis of very strong population and dwelling change forecasting rather than any current day brilliance.

Figure 15 Potential sites in Austral suburb (made by author on MS Powerpoint)

3.4.1 Site A – Tenth Avenue / Edmondson Avenue

Likely to be placed at the intersection of Tenth Avenue and Edmondson Avenue, this is located right beside the athletic field and developing residential land uses. As mentioned earlier, this location depicted moderate coverage in the present-day situation but has got immensely strong potential about its integration with recreational and other future growth fonts.

Figure 16 Site A - Tenth Avenue / Edmondson Avenue (made by author on MS Powerpoint)

3.4.2 Site B – Fifteenth Avenue / Edmondson Avenue

Featured at the intersection of Edmondson Avenue and Fifteenth Avenue, the catchment area covers less number of dwellings than the previous site but considering state government's investment on fifteenth avenue to upscale as fast-paced transport corridor, its future potential holds greater than Site A.

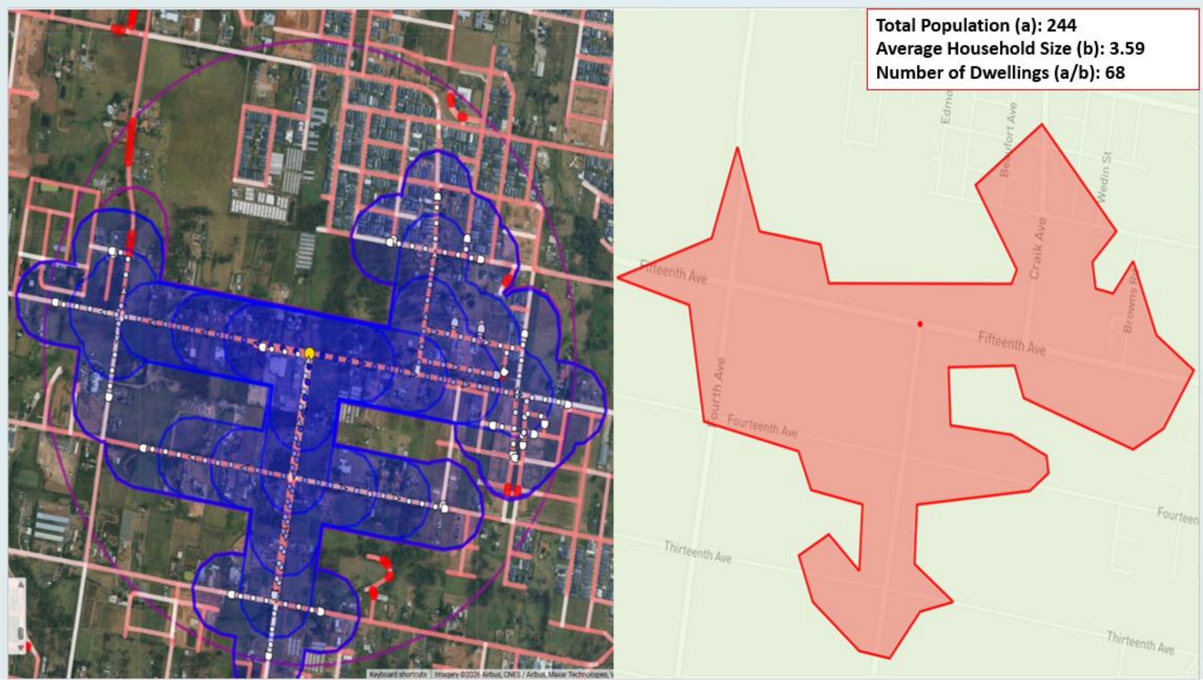

Figure 17 Site B - Fifteenth Avenue / Edmondson Avenue (made by author on MS Powerpoint)

Among the two Austral site locations, Site A currently holds stronger grip on its accessibility catchment and therefore performs more in the favour of walkability assessment against Site B.

3.5 Limitations of Walkability Assessment Tools and Observed Conditions

As brilliant as PedestrianCatch.com and Australian Urban Observatory were in analysing walkable catchments and their accessibility, considerable hindrances bottled up during the procedure.

PedestrianCatch.com primarily assesses the accessibility using street-network geometry and travel time stimulation. While this is extremely useful in enumerating walkable coverage, the tool does not entirely account for physical world experiential dimensions like:

- Availability of shade
- Road crossing safety
- Intensity of traffic
- Continuation of footpaths
- Topography
- Pedestrian's perception of thermal comfort

For instance, certain site locations in Badgerys Creek and Austral technically achieved in 10-minute catchments through the street network but their real on the ground walking environment observed through the satellite imagery appeared fragmented, vehicle-oriented and lack of continuous footpath. Similarly, some Liverpool sites depicted stronger pragmatic walkability because of finer street grids and mixed uses, despite having comparable radial distances.

The Australian Urban Observatory also showcased its share of limitations. In certain neighbourhoods, AUO data coverage appeared inconsistent, particularly within rapidly developing western growth areas and lower-density areas. During the dwelling estimation process, certain selected centre points initially returned population and dwelling values to zero despite visible surrounding structures and road networks. However, when the centre point was slightly shifted within the same precinct, the platform generated valid demographic estimates again. This can imply that certain catchment outputs may be sensitive to:

- Exact centre-point placement
- Mesh-block alignment
- Data aggregation boundaries
- Incomplete spatial coverage

Accordingly, these calculated dwelling estimates are to be interpreted as indicative rather than absolute. Moreover, another important limitation is that walkability assessments mostly reflect current urban conditions and not future scenarios. Areas present in Badgerys Creek and Austral are expected to undergo major transformation with Western Sydney International Airport and the Aerotropolis. Therefore, their walkability assessment are likely to overlook their long-term transport demand.

Despite these limitations, the combined use of:

- PedestrianCatch,
- Australian Urban Observatory
- ArcGIS Pro
- Entropy-weighted overlay analysis
- AHP-based suitability mapping

Provided us with a very useful comparative narration for narrowing broader opportunity zones into more realistic candidate station sites. The walkability findings are summarised below:

Table 14 Comparative narration of Walkability findings

Site	Estimated Dwelling Coverage	Relative Performance
Liverpool C – A34/ Macquarie Street	3343	Strongest Overall Catchment
Liverpool A – Hume Highway/ Southwestern Highway	934	Strong residential integration
Liverpool B – Newbridge Road	364	Moderate catchment: strong distribution logic
Austral A – Tenth Avenue / Edmondson Avenue	82	Moderate future growth and accessibility
Austral B – Fifteenth Avenue / Edmondson Avenue	68	Poor accessibility with promising future
Badgerys Creek A – A9/ Badgerys Creek Road	42	Low current demand with strategic future value
Badgerys Creek B – A9 / Mercy Road	27	Least accessibility

4. NSW Spatial Digital Twin Assessment

One of the shortlisted sites, Liverpool Site C near A34 and the Macquarie street was further examined using the NSW Spatial Digital Twin to better understand its relationship with the surrounding infrastructure, land use patterns, and environmental conditions.

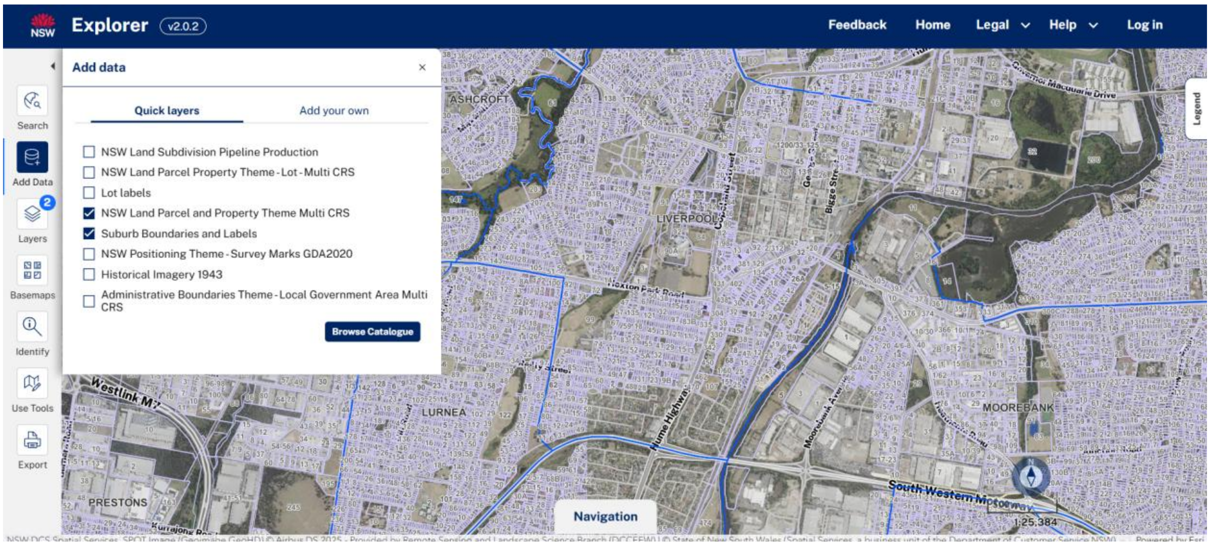

Figure 18 Liverpool LGA in NSW Spatial Digital Twin

The Spatial Digital Twin provided additional layers that were not fully visible within earlier tools like PedestrianCatch, AUO, or Kepler.gl. These include:

- Detailed land use zoning
- Cadastral parcels
- Transport infrastructure networks
- Road hierarchy
- Environmental constraints
- Rail infrastructure

The Digital Twin depicted that the site is strongly integrated within an existing mixed-use urban structure comprising:

- Dense residential neighbourhood's
- Major road corridors
- Nearby rail infrastructure
- Commercial land uses
- Established pedestrian network

However, several limitations were still prevalent. The Digital Twin does not fully capture:

- Pedestrian comfort
- Footpath quality
- Shading
- Traffic safety
- Real-time congestion
- Social perceptions of accessibility

While it visualises existing conditions effectively, it is less capable of representing future behavioral change associated with airport growth or future TOD intensification in the long-term demographic transformation.

Therefore, the Digital Twin was mostly useful as a complementary verification tool rather than a standalone decision-making platform. It helped in validating the physical feasibility and urban integration of the shortlisted site, while the broader analytical justifications continue to rely on the combined outputs from:

- Entropy-weighted suitability analysis
- AHP assessment
- Walkability testing
- Demographic profiling
- Strategic planning assessment

5. Site Comparison and Final Recommendation

5.1 Comparative Site Evaluation

Following the council-level suitability assessment, detailed spatial analysis, and walkability testing, the shortlisted potential sites were comparatively evaluated using all the strategic planning criteria gathered throughout the study.

Table 15 Comparative Site Evaluation

Criteria	Liverpool C – A34 / Macquarie Street	Austral A – Tenth Avenue / Edmondson Avenue	Badgerys Creek A – A9 / Badgerys Creek Road
Council-wide suitability ranking	Highest	Third highest	Second highest
Existing dwelling units covered	3343	82	42
Population within 10-minute catchment	8158	275	134
Existing Public Transport accessibility	Strong	Moderate	Weak
Car dependency reduction potential (number of car-based commuters)	Very high	Moderate	Future-oriented
Future growth potential	High	Very High	Extremely High
Airport/Aerotropolis integration	Moderate	Moderate-High	Very High
Existing urban activity	Strong mixed-use centre	Emerging suburb	Low activity
Environmental/ heritage constraints	Moderate	Moderate	High
Strategic transport integration	Strong	Moderate	Very high long-term potential
Overall short-term viability	Highest	Moderate	Least
Overall long-term value	High	High	Very High

5.2 Final recommendation

The comparison depicts an explicit distinction between sites with strong current urban demand and those with higher strategic growth potential in the future. Based on the combined results from the entire study, the most suitable site recommended for the construction of metro is Liverpool Site C – A34 / Macquarie Street.

This recommendation is primarily supported by its exceptionally large walkable catchment and potential for existing urban integration. The Australian Urban Observatory analysis estimated that approximately 8,158 residents fall within the 10-minute catchment area around the site, covering an approximate 3,343 dwellings. This is substantially higher than all other analysed site locations. So far, the strongest site in Austral covered only 82 dwellings and 42 in case of Badgerys Creek.

This site also strongly relates with the broader insights from the council-wide entropy weighted suitability assessment. Liverpool recorded the highest overall suitability score (0.541) due to:

- The highest population density
- Strongest local employment concentration
- Highest public transport usage
- Highest number of people walking to work
- And the strongest overall commuter activity within the council area.

This location further benefits from its integration within an already established mixed-use urban setting comprising of:

- Residential neighbourhoods
- Employment activity
- Existing transport corridors (close to Liverpool train station)
- Commercial services
- Pedestrian street connectivity

Unlike numerous potential sites within western-growth area, Liverpool already displayed strong TOD readiness and immediate ridership potential. This adds onto attracting substantial amount of capital investment for metro infrastructure, where demand already exists alongside potential for future growth.

Moreover, the ability of the site to complement rather than entirely duplicate the existing Liverpool catchment. The broader catchment area of these two public transport hubs can enable configuration for distributing commuter movement efficiently in the Liverpool urban core, reducing the concentrated pressure at the interchange node. This reflected broader metropolitan transport planning approaches where secondary high-capacity stations support network resilience, as can be seen in the case of Parramatta.

However, the recommendation also involves certain trade-offs, the recommended site is relatively less strategically connected to the Western Sydney International Airport and the Aerotropolis than the Badgerys Creek and even Austral. The recommendation is therefore a stronger priority from the lens of:

- current accessibility
- immediate demand

Rather than purely long-term airport-oriented metropolitan expansion. Additionally, to further improve confidence in the recommendation, various extra datasets can be considered valuable in the future work. These include:

- Projected airport employment forecasts
- Future bus network restructuring plans
- Detailed pedestrian quality and acceptance audits
- Traffic congestion modelling
- Land acquisition feasibility
- Construction cost estimates
- Flood-risk assessments

Ultimately, the evidence backs up that Liverpool Site C proves to possess the strongest balance among:

- Existing demand
- Accessibility
- Transport integration

- Strategic urban functionality

Making it most suitable metro station participant among the analysed sites under present day planning stipulations.

Lastly, this study relied only on static datasets like ABS Census data and land suitability overlays but did not integrate any live IoT data. Involving that could significantly improve the overall quality of study. For instance, including bus passenger loads, parking occupancy and public transport frequency can help verify if the sites are performing to their maximum or not. The same goes for NSW Spatial Digital Twin, which offers more of static spatial representation but no continuous live urban monitoring datasets. Integrating IoT sensor feeds and AI-based predictive modelling into the platform can uplift the overall study.